Eco-evolutionary simulations coupled with spatially-explicit palaeoclimate reconstructions spanning the Americas for the last five million years yielded more South-to-North than North-to-South successful exchanges. These findings reveal that that the asymmetry of the Great American Biotic Interchange (GABI), incontestably North-to-South, may not have emerged from palaeoclimate dynamics alone, challenging classical hypotheses to explain this macroecological enigma.